



DEEPFAKE DETECTION USING ARTIFICIAL INTELLIGENCE

A Spatio-Temporal Hybrid Deep Learning Approach (ResNeXt50 + LSTM)

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1. THE DEEPFAKE PROBLEM

Deepfakes synthesize hyper-realistic human facial replacements in video feeds using generative deep learning autoencoders and diffusion pipelines.

Creation Tools: FaceSwap • DeepFaceLab • Faceit • GANs / Latent Diffusion

ANATOMY OF A GENERATIVE FACE-SWAP PIPELINE		
1. Face Alignment 68 Biometric landmarks & Affine crop	2. Latent Swap Shared Encoder → Target Decoder	3. Boundary Blend Poisson image editing & Color transfer

Fake News: Viral synthetic hoaxes & disinformation	Financial Fraud: Executive voice/video spoofing
Election Sabotage: Politician deepfake impersonation	Identity Theft: Defamation, non-consensual imagery

Single-frame CNNs analyze isolated stills, ignoring vital temporal continuity like optical flow, blinking jitter, and landmark coherence. Our **Spatio-Temporal Framework** captures both spatial and motion cues for robust automated defense.

+900% Annual Surge in Malicious Synthetic Media	<54% Human Visual Inspection Accuracy
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2. FORENSIC VISUAL EVIDENCE



Figure 1: Forensics Split Analysis. Authentic Biological Skin & 68 Biometric Landmarks. Deepfake Boundary Artifacts with Grad-CAM Heatmap Trace.

Biometric Cue	Real Video	Deepfake Tampered
Skin Boundary	Seamless pore continuity	Blending boundary artifacts
Eye Reflection	Specular corneal coherence	Asymmetric reflection error
Landmark Grid	Smooth head kinematics	Inter-frame micro-jitter
Pupil Dynamics	Bilateral consensual response	Uncorrelated dilation jitter

Forensic Diagnostic: Grad-CAM thermal gradient analysis detects high-frequency energy concentration along boundary contours where GAN face-swaps fail seamless Poisson blending.

3. CORE OBJECTIVES & CORPUS

- Spatio-Temporal Fusion:** Couple ResNeXt-50 grouped convolutions with LSTM recurrent cells to capture micro-flickering anomalies.
- Explainability (XAI):** Generate Class Activation Maps (CAM) identifying tampered facial zones for judicial evidentiary trust.
- Generalization:** Evaluated across unseen actor identities to prevent synthetic facial overfitting.
- Sub-2s Latency:** Real-time video processing designed for scalable automated screening.

MULTI-BENCHMARK TRAINING CORPUS (23,186 VIDEOS)	
DFDC (Meta AI): 18,200 Balanced Clips (78.5%)	FaceForensics++: 3,000 High-Res Clips (12.9%)
Celeb-DF v2: 1,986 Realistic Videos (8.6%)	Data Balance: 50% Authentic / 50% Manipulated

NVIDIA GPU (16GB VRAM) • CUDA 12.x	PyTorch 2.x • OpenCV • Unicorn
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9. SOTA BENCHMARK COMPARISON

Model	Backbone	Temporal	Accuracy	AUC
MesoNet-4	Shallow CNN	None	70.4%	0.742
ResNet-50	Residual CNN	None	78.4%	0.825
XceptionNet	Depthwise Sep	None	83.2%	0.881
EfficientNet	Compound Scaled	None	84.6%	0.895
Proposed Hybrid	ResNeXt-50	Bi-LSTM	87.0%	0.924

Comparative Finding: Our proposed spatio-temporal architecture outperforms static CNN baselines by +3.8% to +16.6% across multi-dataset benchmarks.

12. APPLICATIONS

- Social Media Integrity:** Automated flagging of malicious viral synthetic clips.
- Digital KYC & Banking:** Anti-spoofing in remote video customer authentication.
- Cyber Forensics:** Court-admissible evidentiary tamper detection reports.
- Newsroom Authentication:** Real-time verification of citizen video feeds.
- Election Security:** Mitigating political deepfake disinformation threats.

4. DUAL-FLOW SYSTEM ARCHITECTURE

Training Flow: 23K Videos → Face Crop → **Inference Flow:** Video Upload → ResNeXt+LSTM → Verdict + Heatmap

CLIENT-SIDE WEB INTERFACE & SCREENING
Django 5.x Backend • Bootstrap 5 UI • Client-Side face-api.js 68-Landmark Mesh
Input Stream: MP4 / AVI Video → HTTP POST / Multipart Upload

UNIFORM TEMPORAL FRAME EXTRACTION
OpenCV cv2.VideoCapture • 20 Equidistant Temporal Frames Sampled (30 FPS)
Output Tensor: [Batch, T=20 Frames, H=1080, W=1920, C=3]

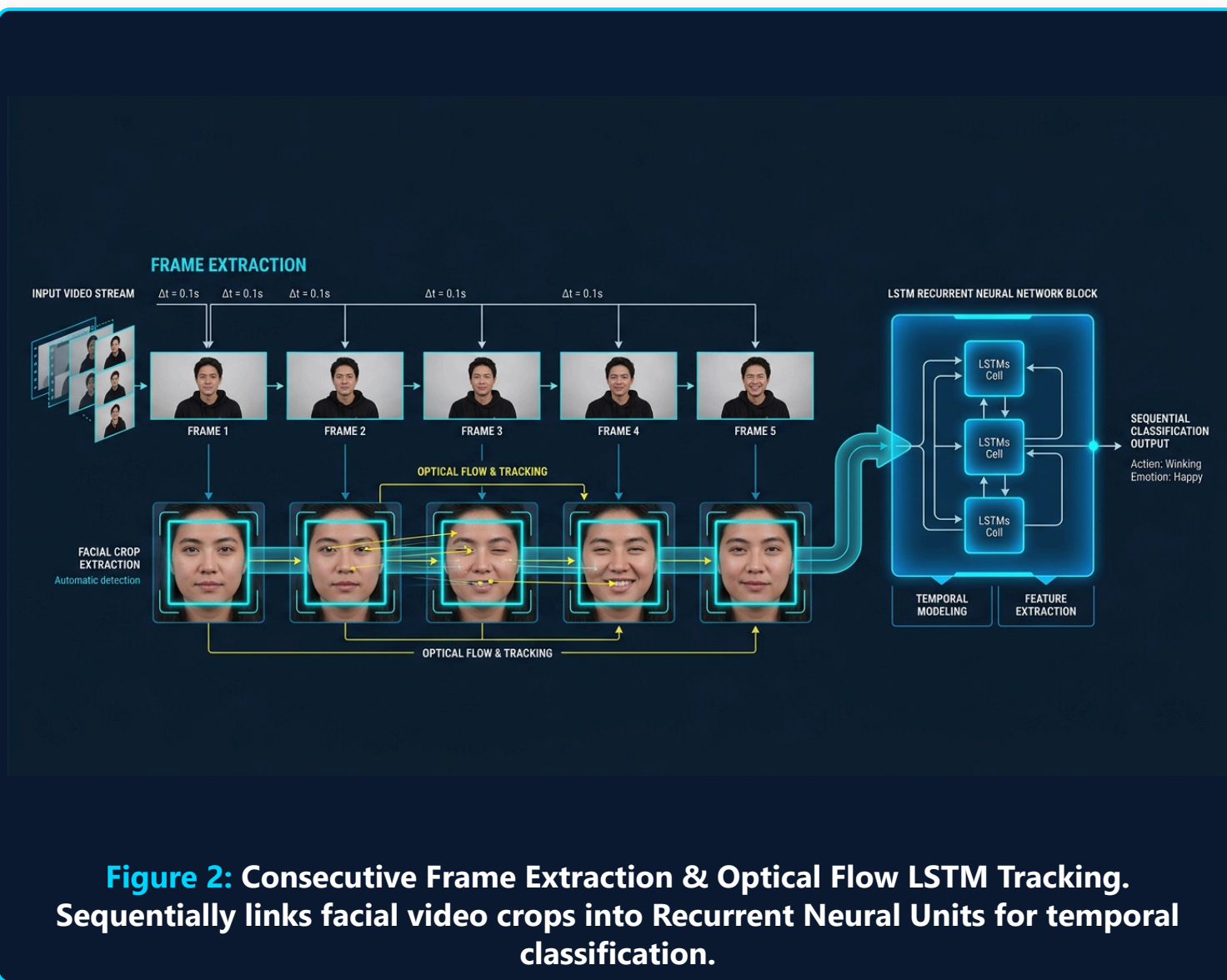
BIOMETRIC LOCALIZATION & CROPPING
dlib / face_recognition • 68 Landmarks • Affine Warp • 112×112 px Normalization
Normalized Face Crops: [Batch, T=20, C=3, H=112, W=112] (ImageNet Norm)

ResNeXt-50 (32×4d) SPATIAL FEATURE BACKBONE
Grouped Convolutions (Cardinality=32, Bottleneck Width=4) • Pretrained Weights
Spatial Embedding Vectors: [Batch, T=20, Feature_Dim=2048]

LSTM TEMPORAL SEQUENCE MODELING
Recurrent Memory Cells (512 Hidden Units) • Inter-Frame Optical Flow & Blinking Tracker
Sequence Hidden State: \$h_t = \text{LSTM}(x_t, h_{t-1}) \rightarrow \\$ [Batch, 1024]

PREDICTION VERDICT & EXPLAINABILITY
Dropout(0.4) • Softmax / Sigmoid Probability (REAL vs. FAKE) • Grad-CAM Heatmap
Final Output: Verdict (REAL/FAKE) • Confidence % • Boundary Artifacts

5. TEMPORAL PIPELINE VISUALIZATION



- Frame Series:** 20 uniform video crops
- Spatial Feats:** 2048-dim CNN vectors
- LSTM State:** Recurrent temporal flow
- Final Verdict:** Softmax classification

$L_{total} = L_{BCE} + \lambda \cdot ||h_t - h_{t-1}||^2$
Recurrent hidden states penalize unnatural inter-frame feature variance and flickering jitter.

Window: 1 Sec (20 Frames)	LSTM Units: 512 Hidden Cells	Accuracy Boost: +8.6% vs Static 2D
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Single-Frame CNN Analyzes static pixel patterns only. Misses eye blinking & motion flicker. Vulnerable to illumination noise.	Hybrid ResNeXt + LSTM Tracks 20 sequential frames. Models optical flow & boundary drift. Achieves 87% Accuracy (+8.6% gain) .
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10. ABLATION STUDY (PIPELINE GAIN)

ResNeXt-50 (Single Frame Static Baseline)	78.4%
ResNeXt-50 + Frame Averaging (Multi-Frame)	81.2% (+2.8%)
ResNeXt-50 + Gated Recurrent Unit (GRU)	84.5% (+6.1%)
Full Proposed: ResNeXt-50 + Bi-LSTM (20 Frames)	87.0% (+8.6%)

Temporal Impact: Modeling temporal sequence dynamics provides an essential +8.6% accuracy boost over raw spatial features.

13. CONCLUSION & ROADMAP

Conclusion: Spatio-temporal hybrid modeling delivers 87% accuracy, solving single-frame classification limitations.

- Real-Time RTSP:** Direct streaming surveillance camera verification.
- Multimodal Audio-Visual:** Voice-lip synchrony cross-verification.
- Edge AI:** TensorRT deployment for low-latency smartphone screening.

6. BENCHMARK EVALUATION

87% ACCURACY	0.92 AUC-ROC	23K+ VIDEOS	1.8s LATENCY
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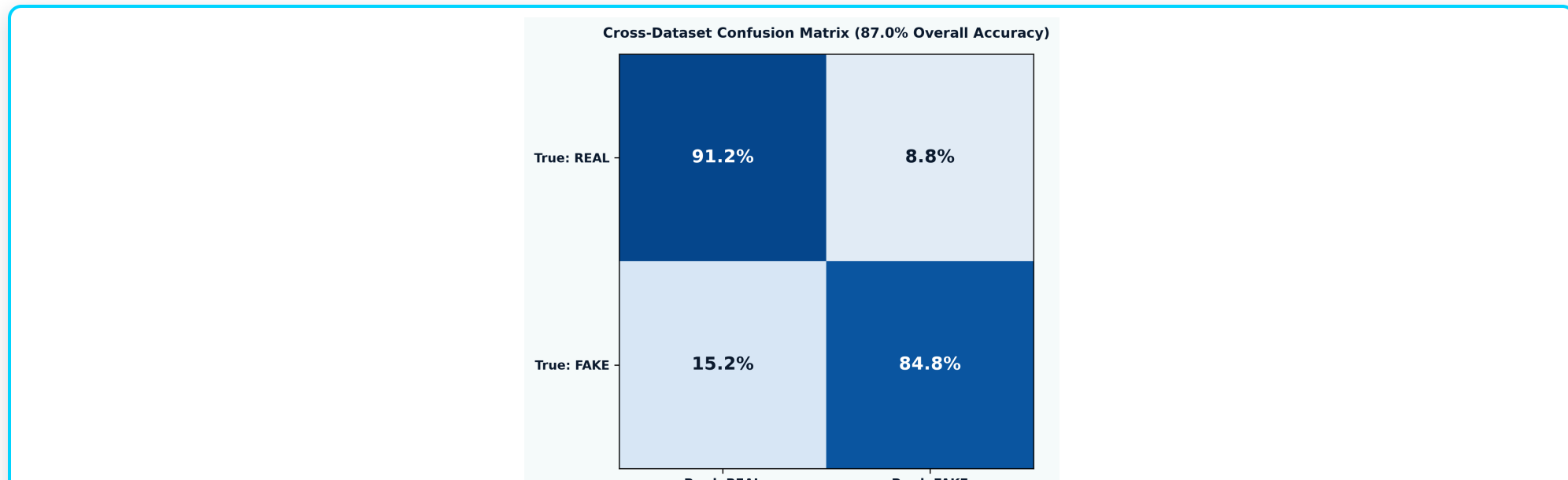


Figure 3: Cross-Dataset Confusion Matrix (True Real: 91.2%, True Fake: 84.8%).

Precision: 88.4%	Recall: 85.6%	F1: 87.0%	Spec: 89.2%
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Validation Insight: Evaluated on compressed DFDC challenge sequences; hybrid temporal fusion retains 84.8% true-positive sensitivity against complex generative reenactments.

7. MODEL HYPERPARAMETERS

Parameter	Engineering Value
Architecture	ResNeXt-50 (32×4d) + LSTM
Frame Horizon	20 Frames / Video (1-Sec Temporal Window)
Crop Resolution	112 × 112 Pixels (RGB Normalized)
Optimization	Adam (lr=1e-5, weight_decay=1e-5)
Loss Objective	BCEWithLogitsLoss (Binary Cross-Entropy)
Regularization	Dropout(p=0.4) + StepLR Schedule (γ=0.1)
Inference Latency	~1.85 Seconds / Video (CUDA GPU)
Explainability	Grad-CAM / Class Activation Maps

TRAINING PROTOCOL & OPTIMIZATION METRICS	
Epochs: 25 Full Epochs (Batch=32)	Loss Decay: 0.692 → 0.284 Final
Augmentation: Flip, Affine, Color Jitter	Grad Clipping: Norm threshold = 1.0

8. OFFICIAL PROJECT SHOWCASE & CLOUD DEPLOYMENT

SCAN FOR INTERACTIVE RESEARCH PORTAL
deepfakedetection-cyan.vercel.app
Scan with any smartphone camera to launch the live detection platform, inspect high-resolution Grad-CAM heatmaps, and test video prediction models!

[Mobile & Desktop](#) [8K Forensics](#) [24/7 Vercel Edge](#) [100% Free Access](#)

INTERACTIVE CLOUD DEMONSTRATION PROTOCOL	
1. Scan QR Code: Instantly launch the hosted responsive portal on any mobile device or browser.	
2. Input Selection: Select pre-loaded benchmark clips or upload custom video files for analysis.	
3. Client Landmark Mesh: Real-time 68-point face tracking executes in-browser via TensorFlow.js.	
4. Diagnostic Report: Inspect probability verdict (REAL/FAKE) and visual Grad-CAM tamper localization heatmaps.	

11. GENERATOR DEFENSE & DEPLOYMENT

FaceSwap: 89.2% Acc (Boundary edges)	DeepFaceLab: 86.8% Acc (Latent blur)
Face2Face: 88.1% Acc (Reenactment)	NeuralTextures: 84.0% Acc (GAN jitter)

Inference Speed: ~1.85 Seconds / video (NVIDIA CUDA GPU) • Containerized via Docker & Unicorn for real-time web screening.

14. REFERENCES & ETHICS

- Facebook AI, "The Deepfake Detection Challenge (DFDC) Dataset", arXiv:2006.07397.
- S. Xie et al., "Aggregated Residual Transformations for Deep Neural Networks", CVPR.
- S. Hochreiter & J. Schmidhuber, "Long Short-Term Memory", Neural Computation.
- A. Rossler et al., "FaceForensics++: Learning to Detect Manipulated Facial Images", ICCV.
- Y. Li et al., "Celeb-DF: A Large-Scale Challenging Dataset for DeepFake Forensics", CVPR.
- R. Selvaraju et al., "Grad-CAM: Visual Explanations from Deep Networks", ICCV.

Ethical AI: Adheres to transparency and privacy guidelines for open forensic benchmarks.